AKSHAY MHATRE

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Summary

Full-Stack Software Engineer with 7.5+ years of experience building production systems across React/Next.js, Vue.js, TypeScript, Node.js, event-driven microservices, authentication/SSO, micro-frontends, and distributed application debugging. Strong in frontend/backend integration, RBAC/security flows, production troubleshooting, and AI/security research publications.

PROFESSIONAL WORK EXPERIENCE

Texas A&M University – Central Texas | Graduate Research Assistant — Killeen, Texas | **May 2025 - May 2026**

- Improved a privacy-aware model de-duplication approach across ResNet, ViT, RoBERTa, and multimodal models by designing a Top-K high-accuracy block-selection strategy from 100+ experimental runs for an ASU-led journal paper. Also, ranked model metrics summaries based on MIMIC datasets, validation results and findings on federated learning in Python.
- Presented research work on comparing LLM-based bug detection and repair capabilities across production libraries such as OpenSSL, pandas, & NumPy at IEEE Southeast Conference 2026 in Huntsville, AL, with Los Alamos National Lab scientist.
- Conducting ongoing research on defensive Agentic AI using Azure AI Foundry with React as frontend, with emphasis on policy-guardrails (using OPA), anomaly detection, and compromise prevention in autonomous tool-using systems. Building agentic loop & defenses.

WebMD | MH Sub I, LLC — Mumbai, India

Software Engineer I | **Projects: Medscape (.co.uk, .com), Univadis, Helios Platform, WebMD** — **August 2022 - August 2024**

- Developed 50+ responsive features and high-performant reusable state-managed components across server-side rendering (SSR) based applications using React Server Components/Next.js, MUI, and Vue 3, including tables, charts, auto-scroll pagination and user actions tracking modules for displaying and injecting Ads deployed by Adobe launch.
- Migrated from Vue 2 to Vue 3 by implementing custom functionality for deprecated features, state-management from vuex to Pinia, also resolved and accelerated major memory leak issue's identification process by simulating prod env locally using Docker, Kubernetes, Clinic JS and Autocannon eliminating major blocker for multiple teams and their micro-frontends.
- Engineered secure route guards by tying React Router with TanStack Query based cached identity states to router loaders, preventing flicker and client-side authorization bypasses across 5 react/next js full-stack applications.
- Boosted Core Web Vitals scores by 35% by implementing strategic data prefetching via render-phase hooks and component-level code splitting and lazy-loading non-crucial components.
- Managed release workflows and automated Semantic versioning in GitLab pipelines for shared React elements, boosting deliverables by 25%. Also, led debugging of multi-stage Docker images to resolve logging issues and missing modules.
- Designed and optimized Apache Solr collections, schemas, indexing pipelines, and Lucene-based queries for CMS content delivered through RabbitMQ, then exposed and consumed search data through Node.js and Spring REST APIs

Iorta technology solutions pvt ltd — Mumbai, India

Senior Software Engineer - **Frontend Tech-Lead** | **Project:SalesDrive, Tata-AIG Claims System, Banca** — **August 2017 - August 2022**

- Led, managed and trained a team of 10+ engineers on React & Vue.js, Vuetify, REST API integration, Node.js, react native and full end-to-end ownership and delivery of modules for a large-scale AIG claims enterprise system project having 22+ modules.
- Designed an enterprise RBAC system using React Higher-Order Components (HOCs) and render prop patterns, seamlessly blocking unauthorized data mutations and UI state access at granular level based on JWT authorization claims & permission attributes.
- Integrated React micro-frontends with Node.js/Express backend services, AWS Cognito, AWS Lambda, and third-party REST APIs by building a reusable TypeScript Axios client layer with JWT propagation, interceptor-based request handling, typed response models, RBAC claim validation, and centralized API error normalization; supported backend auth flows using Express middleware, Cognito JWT verification, CORS, Helmet, logging, validation, and AWS SDK/Lambda integrations.
- Integrated & worked with multiple interdependent REST API's with Axios, leveraged a serverless managed solution by AWS Cognito, AWS Lambda functions to develop an OAuth SSO (single-sign-on) solution between multiple React frontends, microservices and third-party API services containing millions of records, insurance policy datasets, role mappings in an immutable state-managed micro-frontends based on Redux Toolkit and Vuex respectively.

- Integrated REST APIs and GraphQL services using Axios and Fetch API to connect frontend applications with backend services. And used optimization techniques like code splitting with webpack cache-groups to manage bundle sizes.
- Developed multiple reusables, scalable and unit-tested components (following the TDD approach) with Jest + React Testing Library for React components, also Vue test utils, reducing developer-end testing time & bugs/defects by 50%.
- Collaborated with UI/UX designers, back-end developers, business analysts and client-side stakeholders to understand and build advanced global state-managed features such as dynamically animated policy Benefit Illustration, advisor-customer greetings creator and image editor with *Fabric JS, Chart JS, Three JS* to meet business requirements.

TECHNICAL SKILLS

Programming Languages: JavaScript (Advanced), TypeScript, Java (Intermediate), Python (Intermediate)
Frameworks: React.JS, Redux Toolkit, Vue.js (SSR), Nuxt.js, Vuex, Pinia, Node.js, Next.JS, React Router, TanStack Query, Express.js, Vuetify, Spring Boot, Apache Solr, Tailwind CSS, Sass, SCSS (LESS), HTML, Axios, Scikit-Learn, PyTorch, React Native, Weex
Database Management: MongoDB, PostgreSQL, Redis, MySQL
Software / Tools: LangGraph, Azure AI foundry, Cursor IDE, Jira, REST API integration, Postman, NGINX, GitLab CI/CD, , webpack-bundle-analyzer, Clinic.js, Jest, Vue test-utils (unit testing), Docker, Kubernetes, Rancher, Confluence, DevOps, Kanban, Agile Methodologies, Scrum, YouTrack, WebGL, html CANVAS, RAGFlow, Codex

EDUCATION

Texas A&M University - Central Texas, Department of Computer Information Systems Killeen, Texas
MS in Information Systems | **Concentration**: Applied ML, Data Analytics | **GPA**: 3.87 **May 2026**

University of Mumbai, Universal College of Engineering Mumbai, India
Bachelor of Engineering in Information Technology | **Concentration**: Software Engineering, Computer Graphics **July 2017**